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PORTFOLIO

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SMARTWATCH PCB PROJECT

Project Goal

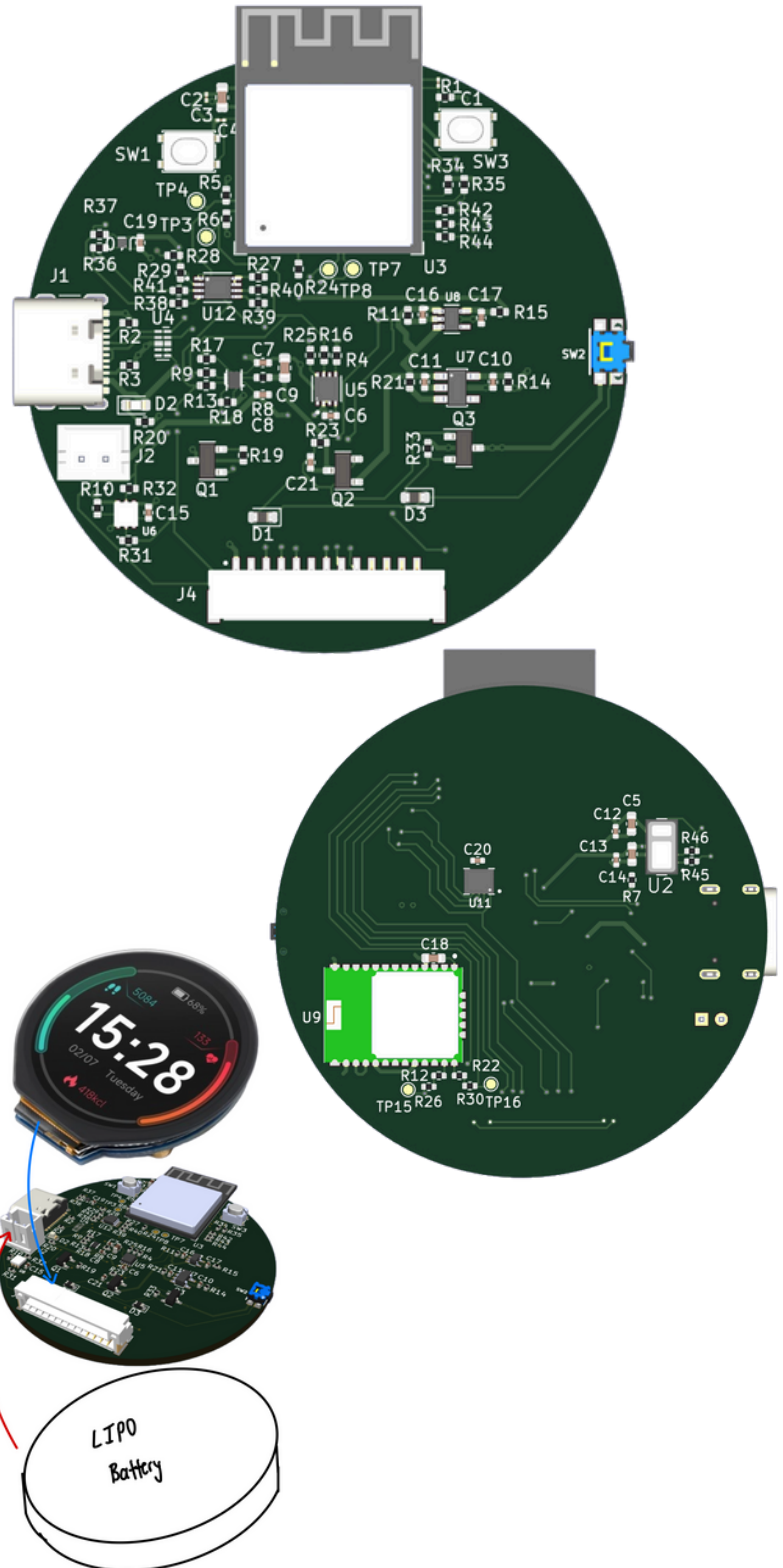
- Design and develop a compact, low-power smartwatch integrating multiple sensors and wireless connectivity for health monitoring and activity tracking, with efficient power management and implemented with an 1.28" circular LCD

Key Skills

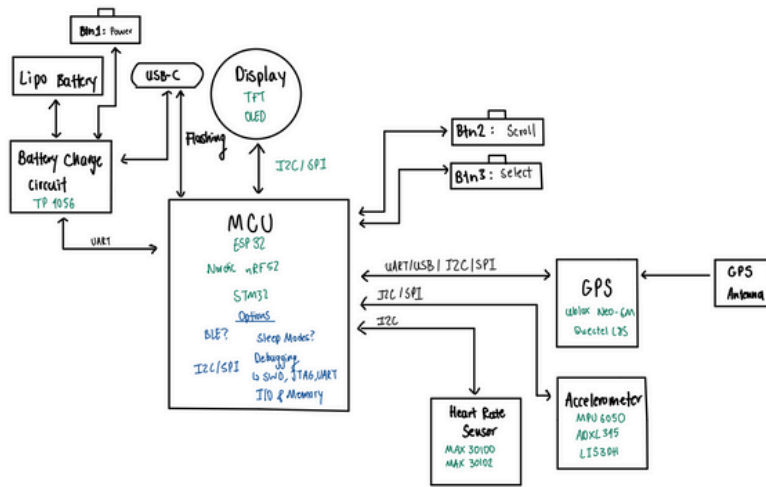
- **PCB Design (KiCAD):** 4-layer board layout in KiCAD with signal integrity, power distribution, and compact routing
- **Sensor Integration:** I²C and SPI communication with IMU, heart rate/SpO₂ sensor, GPS, compass, and ambient light sensors
- **Power Management:** USB-C charging, soft-latch power control, battery fuel gauging, and multiple voltage rails (1.8 V, 3.3 V)
- **Hardware Validation:** Bus topology design, level shifting, testpoint planning, and interface debugging
- **Protocols & Interfaces:** USB, I²C, SPI, UART, BLE, Wi-Fi, GNSS

Project Features

- Designed a 1.28" circular 4-layer PCB integrating the **ESP32-S3-MINI-1** with Wi-Fi, BLE, and USB support
- Integrated sensors via I²C/SPI for heart rate (MAX30102), IMU (LSM6DSOX), magnetometer (AK0991x), GPS (MAX-M10S / CAM-M8Q), and ambient light monitoring, with appropriate **level shifting** for 1.8 V devices
- Implemented **USB-C** power input with battery charging, **power path** management, fuel gauge, and **soft-latch** power control circuit, LDOs for regulated 3.3 V and 1.8 V power rails
- Connected a 1.28" SPI TFT LCD with capacitive touch interface for user interaction
- Focused on signal integrity, proper **test point** and **jumper** placement, and **controlled impedance** routing for high-speed and RF lines



Initial System Diagram



I2C Pull-Up Resistor Calculation/Bus Design

IC	Addr	Vol	VOH	t _{rise}
3.3V { VEMLE6030	0x10	0.4	1.3	1000 ns
MAX17049G	0x34 (7bit)	0.5	1.4	1000 ns
0x26198	0x68 (7bit)	0.4	1.3	1000 ns
0x04 (7bit)				
GC9A01 (10b)	0x15 (7bit)	0.6	1.4	1000 ns
1.8V { AK09418	0x0C (7bit)	0.3	-	1000 ns
MAX30102	0x57 (7bit)	0.4	-	1000 ns

$I_{OL} = 3mA$

3.3V: $R_{pullup\ min} = \frac{V_{CC} - V_{OL(max)}}{I_{OL}} = \frac{3.3V - 0.4}{0.003} = 900\ \Omega$

1.8V: $R_{pullup\ min} = \frac{1.8V - 0.4}{0.003} = 467\ \Omega$

$C_b = 15pF \times 4\ devices + 20pF + 20pF$

3.3V: $R_{pullup\ max} = \frac{t_{rise}}{0.8473 \times C_b} = \frac{1\ \mu s}{0.8473 \times 100pF} = 29.8K$

$R_{pullup\ max} = 11.8K$

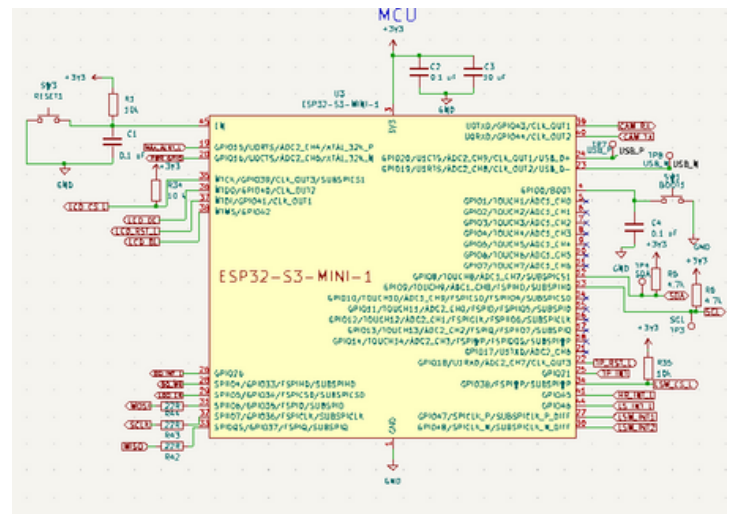
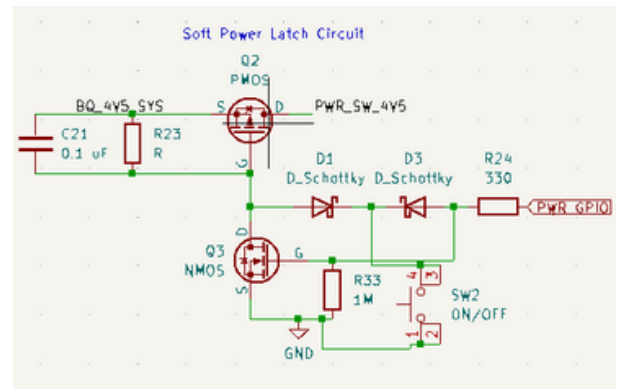
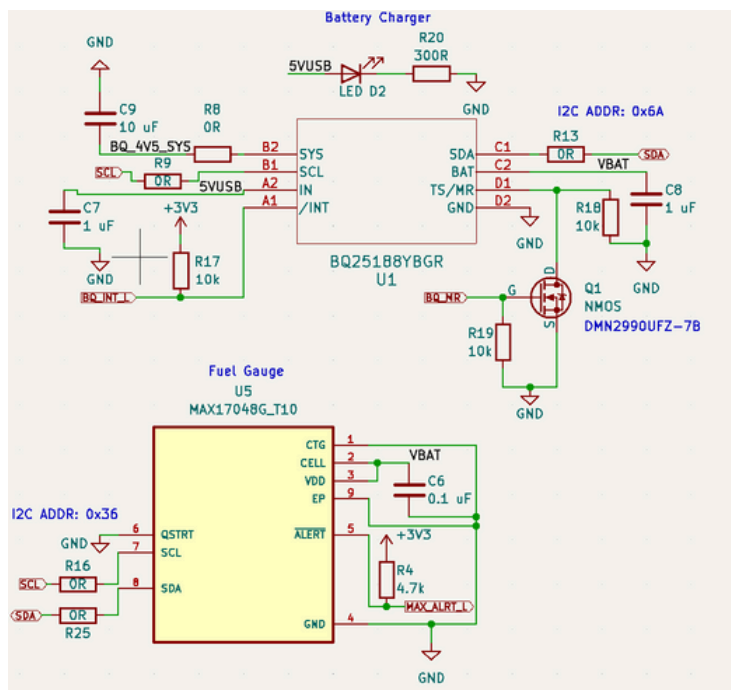
3V3 LDO Junction Temperature Calculation

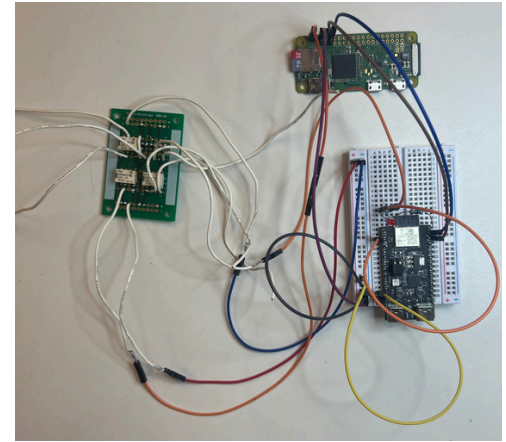
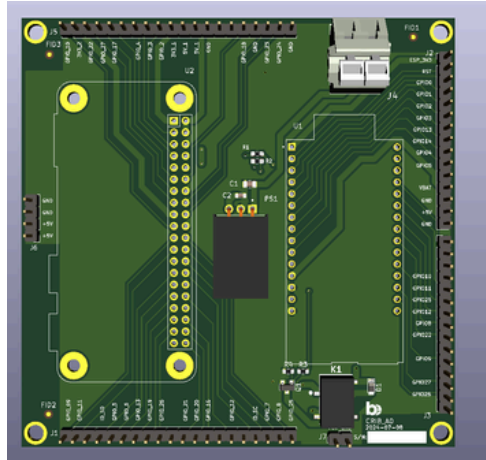
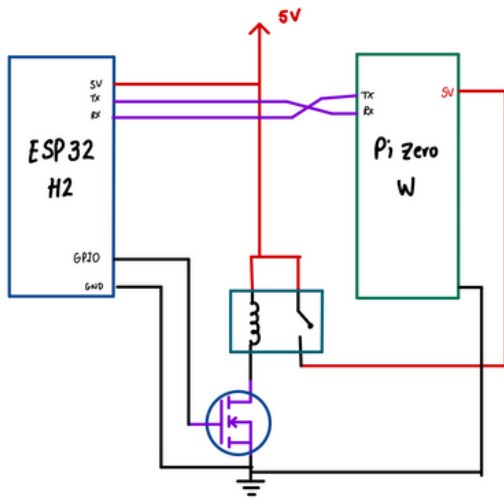
$$\begin{aligned}
 V_{in} &= 4.5V \\
 V_{out} &= 3.3V \\
 I_{load} &= 0.6A \\
 I_{GND} &= 8 \times 10^{-6}A \\
 P_{D\ max} &= (4.5V - 3.3V) \times 0.6A + 4.5V \times 8 \times 10^{-6}A \\
 &= 0.72W + 3.6 \times 10^{-4} \\
 P_{D\ max} &= 0.72036W \\
 \text{Thermal Resistance (Junction to case)} &: 96^{\circ}C/W \\
 \text{Junction temp} &= 0.72036W \times 96^{\circ}C/W \\
 &= 69.15^{\circ}C + 25C \\
 &= 94.15^{\circ}C \checkmark
 \end{aligned}$$

1V8 LDO Junction Temperature Calculation

$$\begin{aligned}
 V_{in} &= 4.5V \\
 V_{out} &= 1.8V \\
 I_{load} &= 0.0012A \\
 I_{GND} &= 3.9 \times 10^{-6}A \\
 P_{D\ max} &= (4.5V - 1.8V) \times 0.0012A + 4.5V \times 3.9 \times 10^{-6}A \\
 &= 3.4155 \times 10^{-3}W \\
 \text{Junction thermal resistance} &: 256.5^{\circ}C/W \\
 \text{Junction temp} &= 256.5^{\circ}C/W \times 3.4155 \times 10^{-3}W + 25C \\
 &= 0.876C + 25C \\
 &= 25.876C \checkmark
 \end{aligned}$$

[VIEW SCHEMATIC PDF](#)





Project Goal

- Reduce the sleep current of the Pi Zero W from current 180 mA for battery powered applications
- Design for future expansion such as adding sensors, support for other MCUs
- Track time spent asleep during operation

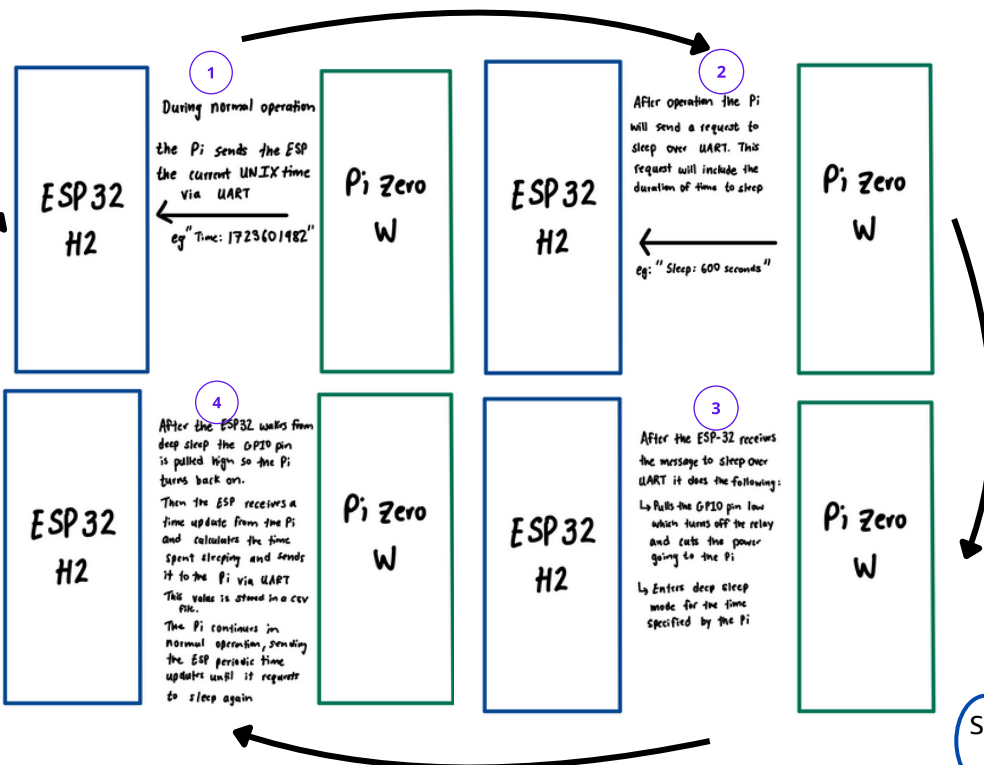
Key Skills

- PCB Design (KiCAD)
- ESP32
- UART (Software & Elec)
- C Programming
- FreeRTOS
- RTC
- Raspberry Pi
- Python

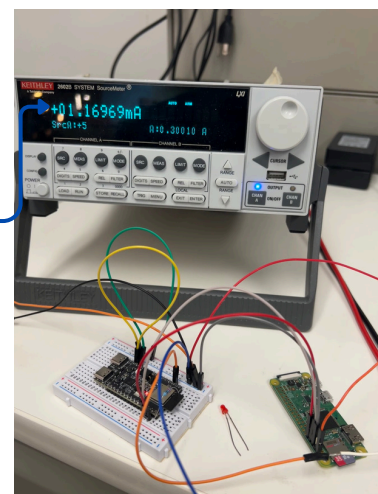
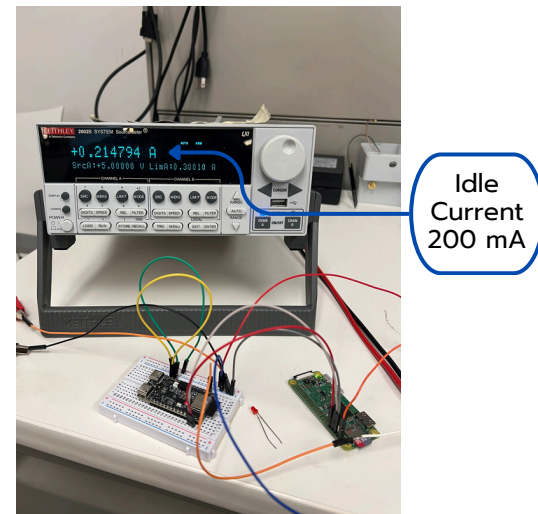
Project Features

- **UART** interface between ESP and Pi Zero
- Pi power controlled by Signal Relay
- Relay controlled by **MOSFET** and ESP GPIO
- On board 24V to 5V DC Converter
- Breakout Headers for expansion
- V2 PCB features support for Adafruit M0
- Sleep Current of **1 mA**

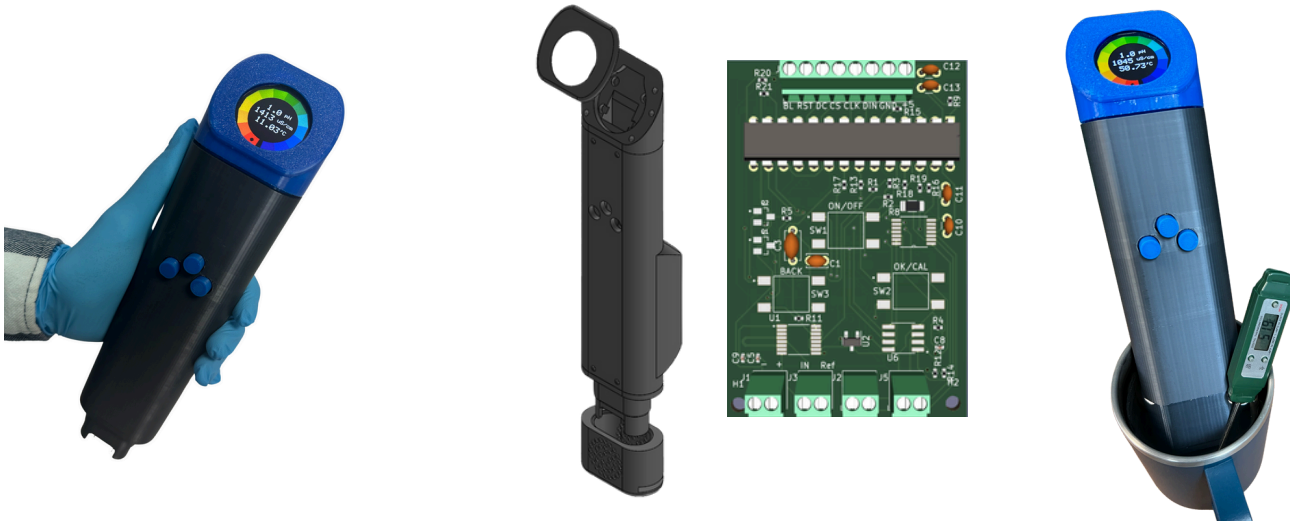
Sequence of Operation



Idle vs Deep Sleep Current



MULTI-PARAMETER WATER TESTER



Project Goal

- Design a handheld device capable of measuring the parameters of a water sample such as PH, Temperature, Conductivity and ORP
- Modular design such that it can be interchanged for different applications and parameters

Key Skills

- Low-Voltage PCB Design (KiCAD)
- High Resolution Sensor Measurement (ADC)
- Signal Amplification (Op-Amp)
- Digital Circuits (Latch Circuit)
- SPI (LCD)
- Fusion360
- C++
- ATMega328

Project Features

- 2 Layer PCB to interface sensors with MCU and LCD
- 12 Bit 4 channel ADC for increased resolution
- Power latch circuit for auto-shutoff
- Op-amp circuit for PH Sensor amplification
- Custom GUI to display measurements
- 3D printed enclosure for prototype

Version 1



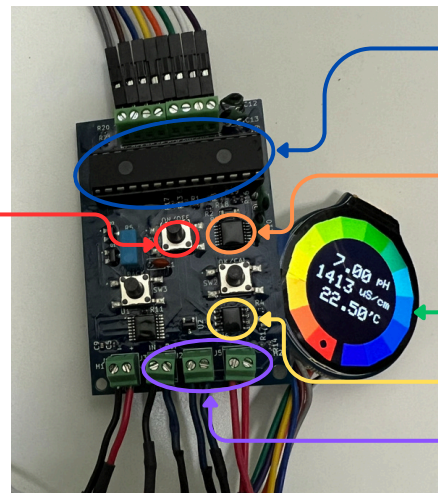
Increased Resolution with ADC



Shrunk PCB size with improved routing and SMT components

Power Button (Latch)

Version 2



Microcontroller

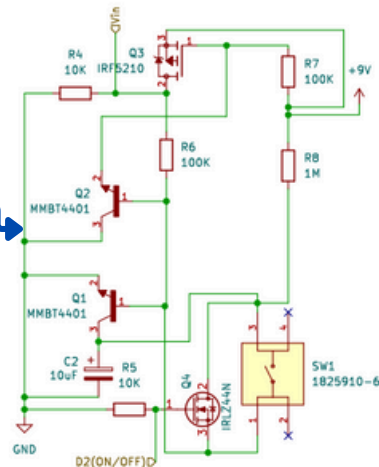
Analog-to-Digital Converter

LCD Display

Op-Amp

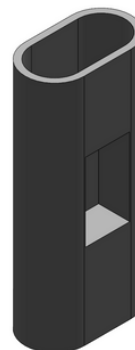
Sensors

Power Latch Circuit

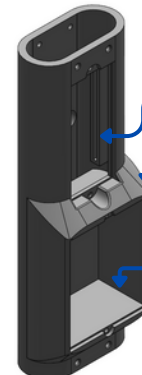


Designed Power Latching Circuit with power button that allowed for timed auto-shutoff of device, as a result saved battery usage

Version 1



Version 2

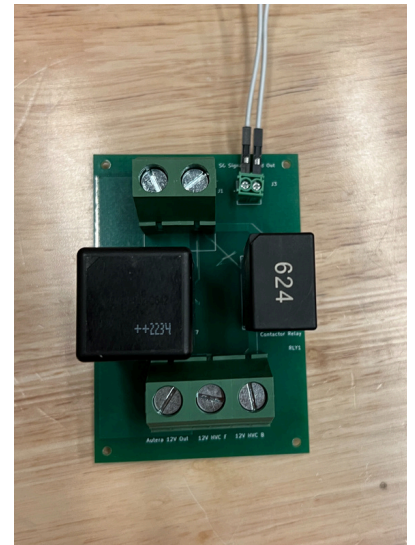
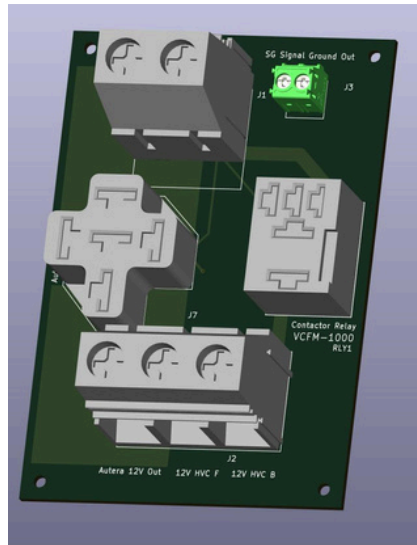
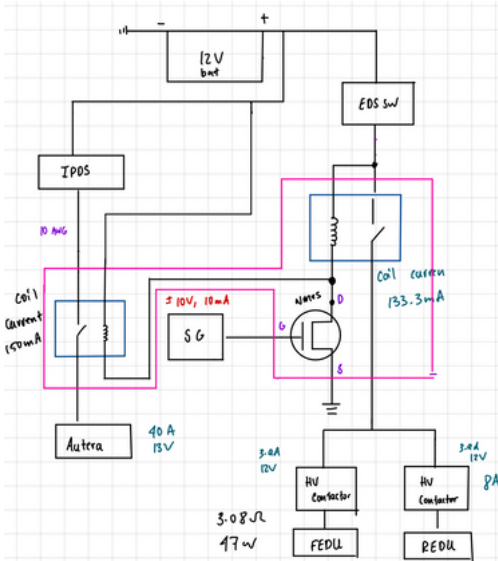


Separate access to PCB for maintenance

Ergonomic and better fit in hand

Changed opening from 9V battery to 6xAAA for increased current capacity

HV CONTACTOR PCB LWAFT



Project Goal

- Design PCB capable of switching the power to the Autera and HV Contactors based on 3.3V signal from Speedgoat
- Be able to handle the max current draw of 40A and 8A respectively

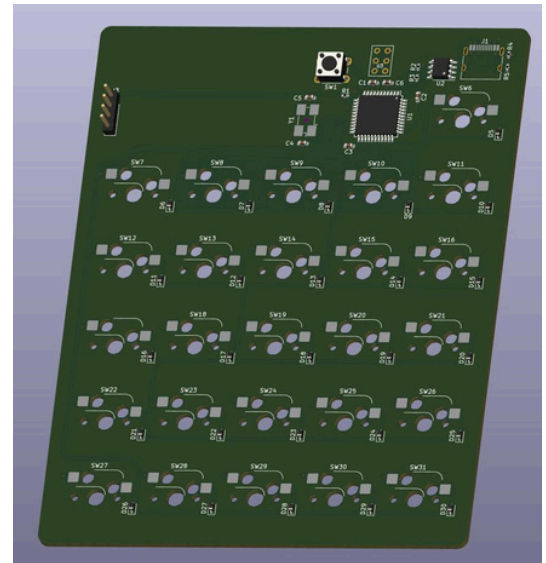
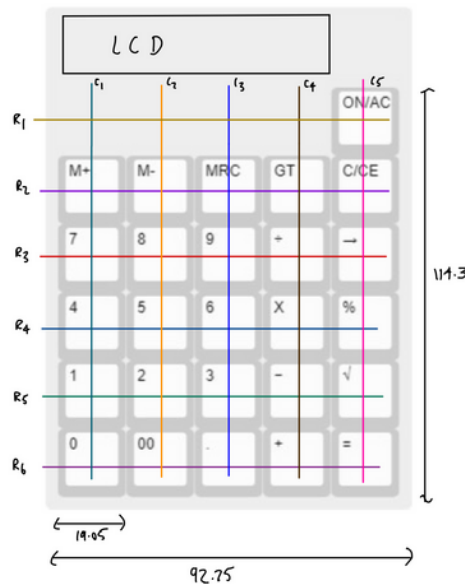
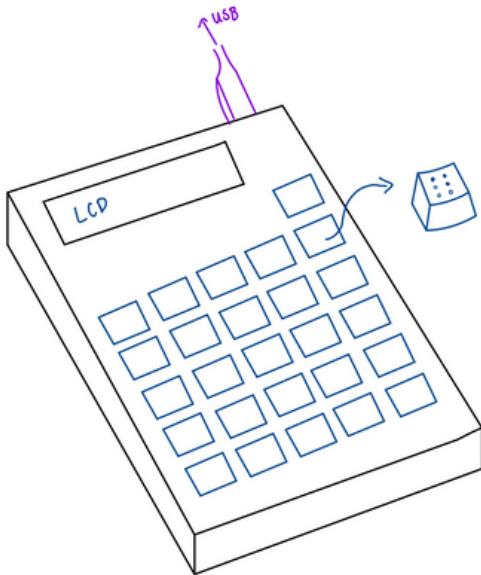
Key Skills

- High Current PCB Design (KiCAD)
- High Capacity Power Relay
- MOSFET Relay Control

Project Features

- Automotive grade Relays to control Power going to Autera and Contactors
- Large terminal blocks for 10 AWG wire
- nMOS to control relays from Speedgoat input

BRAILLE KEYBOARD



Project Goal

- Design a calculator interface that allows users with visual impairment to perform basic calculations and have audible feedback
- Utilize components such as mechanical switches and LCD that are already on hand

Key Skills

- PCB Design (KiCAD)
- ATmega MCU
- I2C
- USB
- Python

Project Features

- ATmega32U4 for detecting input from keys and performing arithmetic
- Diode array for preventing key ghosting
- ISP header as alternate flashing method
- USB connection for power and serial communication with computer
- I2C LCD for debugging purposes